



9314 - Chandra and JWST Observations of NGC 6440

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NGC 6440 F187N	NIRCam Imaging	(1) NGC-6440
	2	NGC 6440 F182M	NIRCam Imaging	(1) NGC-6440

ABSTRACT

Because of their stellar encounters rates, globular clusters (GCs) are the most efficient "factories" in the Universe for the production of exotic, close binary systems; many studies independently predict hundreds of cataclysmic variables in every rich GC. However, this population has remained elusive observationally. Low-luminosity X-ray sources in Galactic GCs, discovered in great numbers with Chandra, are a heterogeneous mix of close binaries, which can only be classified with additional information from other wavebands. We propose to obtain deep Chandra imaging and narrow-band JWST imaging to discover and classify close binaries in one of the richest (yet understudied) globular clusters — NGC 6440.

Chandra Proposal 27300097

OBSERVING DESCRIPTION

JWST Proposal 9314 (Created: Wednesday, August 27, 2025, 12:00:20PM Eastern Standard Time) - Overview

We request NIRCам observations of the globular cluster NGC 6440 using short wavelength filters F187N and F182W and long wavelength filters F405N+F444W and F410M. We will use both modules with the full subarray with target placement on the B4.

For NGC 6440, the two filters will be done in different observations to reduce data volume. The F187N / F405N+F444W observation will use INTRAMODULEBOX 9 and SMALL-GRID-DITHER 4, with MEDIUM8 readout, 1 integration/exposure, and 10 groups/int. The F182M / F401M observation will use INTRAMODULEBOX 6 and SMALL-GRID-DITHER 2, with MEDIUM8 readout, 1 integration/exposure, and 9 groups/int.

Proposal 9314 - Targets - Chandra and JWST Observations of NGC 6440

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC-6440	RA: 17 48 52.6700 (267.2194583d) Dec: -20 21 36.90 (-20.36025d) Equinox: J2000	Proper Motion RA: -1.187 mas/yr Proper Motion Dec: -4.020000051241368 mas/yr Epoch of Position: 2000	
Comments: Category=Stellar Cluster Description=[Globular star clusters]					

Proposal 9314 - Observation 1 - Chandra and JWST Observations of NGC 6440

Observation	Proposal 9314, Observation 1: NGC 6440 F187N								Wed Aug 27 17:00:21 GMT 2025
	Diagnostic Status: Warning								
	Observing Template: NIRCам Imaging								
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(1)	NGC-6440	RA: 17 48 52.6700 (267.2194583d) Dec: -20 21 36.90 (-20.36025d) Equinox: J2000		Proper Motion RA: -1.187 mas/yr Proper Motion Dec: -4.020000051241368 mas/yr Epoch of Position: 2000				
	Comments: Category=Stellar Cluster Description=[Globular star clusters]								
Template	Module		Subarray			Target Placement			
	ALL		FULL			Module B (B4 corner)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size	Subpixel Positions
	1	INTRAMODULEBOX		9		SMALL-GRID-DITHER			4
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time
	1	F187N	F405N+F444W	MEDIUM8	10	1	36	36	37879.325
Special Requirements									
	Fiducial Point Override NRCBS_FULLL								

Proposal 9314 - Observation 2 - Chandra and JWST Observations of NGC 6440

Observation	Proposal 9314, Observation 2: NGC 6440 F182M Diagnostic Status: Warning Observing Template: NIRCam Imaging									Wed Aug 27 17:00:21 GMT 2025
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	NGC-6440	RA: 17 48 52.6700 (267.2194583d) Dec: -20 21 36.90 (-20.36025d) Equinox: J2000			Proper Motion RA: -1.187 mas/yr Proper Motion Dec: -4.020000051241368 mas/yr Epoch of Position: 2000				
	Comments: Category=Stellar Cluster Description=[Globular star clusters]									
Template	Module			Subarray			Target Placement			
	ALL			FULL			Module B (B4 corner)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX		6		SMALL-GRID-DITHER				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F182M	F410M	MEDIUM8	9	1	12	12	11338.029	174661.49
Special Requirements	Fiducial Point Override NRCBS_FULL									